

Engineering Reboot 2026

Week 05 • Professional Python Foundations

Mission: Build software that is reliable, maintainable, and resilient through professional engineering practices.

Day	Focus	Mini Project	Status
21	Exception Handling	Safe Calculator	■
22	Input Validation & Defensive Programming	Validated Registration System	■
23	Professional Function Design	Utility Library	■
24	Type Hints & Modern Python	Type-Hinted Utility Toolkit	■
25	Program Organization	Professional Console Application	■

Weekly Integration Project: Robust Student Management System

Features: multi-module architecture • exception handling • input validation • defensive programming • type hints • persistent storage • professional menu system

Daily Checklist: Notes ■ Journal ■ Exercises ■ Mini Project ■ Validation ■ Screenshots ■ Git Commit ■ Git Push ■

Engineering Principles: Validate input • Fail gracefully • Write reusable functions • Keep functions focused • Use type hints • Organize code professionally